

Manual
Basic Electronics Lab
25B17EC171
Odd Semester 2025

Department of Electronics and Communication Engineering



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List of Experiments

1. Introduction to various components (Resistor, Capacitor, Inductor, and IC) and instruments Multimeter, Breadboard, Regulated D.C. power supply.
2. Introduction to Function generator, Digital Storage Oscilloscope (DSO). Study the properties of sinusoidal, triangular and rectangular pulse signals on DSO.
3. Verification of Ohm's law, KCL and KVL.
4. Verification of Superposition Theorem.
5. Verification of Thevenin's Theorem.
6. To study the forward bias I-V (current-voltage) characteristics of a simple p-n junction diode. Also to determine the forward resistance of the diode
7. To observe the output waveform of half/full wave rectifier and calculate its ripple factor and efficiency
8. To study the reverse bias I-V (current-voltage) characteristics of a Zener diode. Also determine the breakdown voltage, static and dynamic resistances.
9. Realization of desired wave shapes using clipper and clamper circuits.
10. To plot input and output characteristics of a common emitter NPN BJT.
11. To realize inverting and non inverting amplifier configuration using Op-Amp IC- 741.
12. To realize adder and subtractor circuits using Op-Amp IC-741.
13. VIRTUAL LAB 1: BJT Common Base Characteristics. <https://be-iitkgp.vlabs.ac.in/exp/common-base-characteristics/>.
14. VIRTUAL LAB 2: Study of basic properties of operational amplifier: inverting and non-inverting amplifiers. <https://be-iitkgp.vlabs.ac.in/exp/non-inverting-amplifiers/>

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Experiment Number 1

Introduction to various components

1.1 Aim:

To familiarize the students with basic circuit elements such as resistors, inductors, capacitors, ICs, Breadboard and instruments such as Multimeter, Regulated D.C. power supply.

1.2 Equipment/Components required

Resistors (4 band and 5 band resistors of each one), Capacitors (Ceramic disc capacitors with 3-digit codes and the electrolytic type), Inductance box, Multimeter, Breadboard, DC regulated power supply, Connecting wires

1.3 Introduction

1.3.1 Resistor

A resistor is an electronic component that limits or regulates the flow of electrical current in a circuit. It does this by introducing resistance, measured in ohms, which opposes the movement of electrons. Resistors are crucial for controlling current, dividing voltage, and shaping signals in various electronic devices.

The colour coding of the resistors are shown in Fig. 1.3.1.

Resistor Color Code - 4 Band, 5 Band and 6 Band

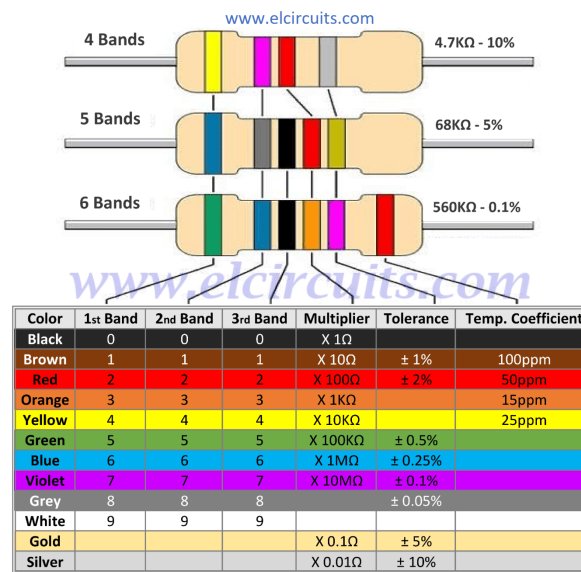


Figure 1.3.1: Color Coding of resistors

1.3.2 Capacitor

A capacitor is an electronic component that stores electrical energy in an electric field. It essentially acts like a temporary "charge reservoir" in a circuit, capable of accumulating and releasing electrical energy. Capacitors are widely used in electronic devices for various purposes, including energy storage, filtering, and signal coupling.

The numeric coding of the capacitor is shown in Fig. 1.3.2.

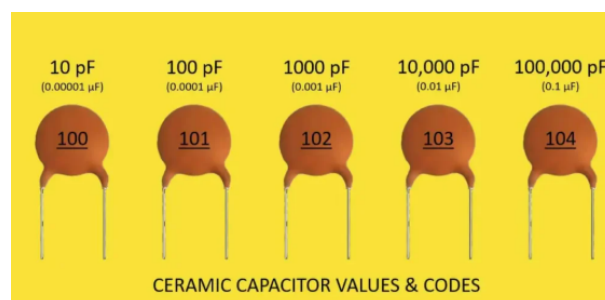


Figure 1.3.2: Numeric Coding of ceramic capacitor

1.3.3

1.3.3 Inductor

An inductor is a passive electrical component that stores energy in a magnetic field when electric current flows through it. Essentially, it's a coil of wire, often wrapped around a core (like iron or ferrite) to enhance its magnetic properties. When current flows through the coil,

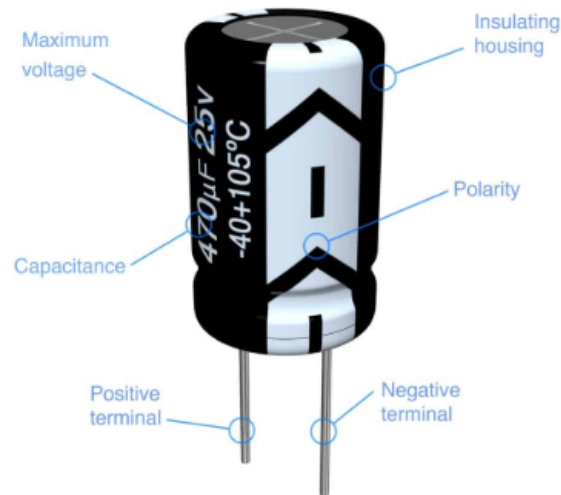


Figure 1.3.3: Electrolytic Capacitor

a magnetic field is generated around it, and this field stores energy. Inductors are used to control or filter electrical signals and are essential in various electronic circuits and devices.

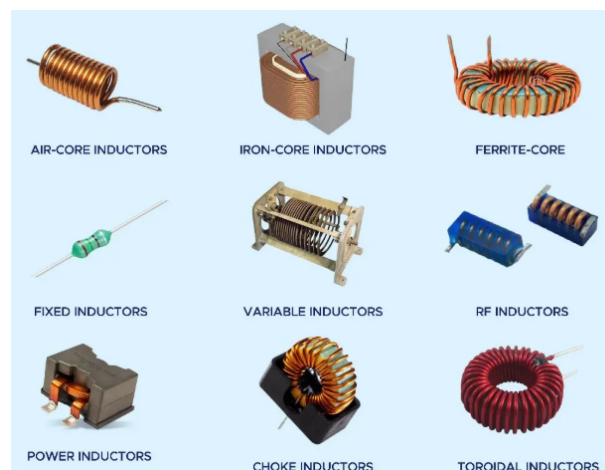


Figure 1.3.4: Different types of inductors

1.3.4 Integrated Circuits

An integrated circuit (IC), also known as a microchip or chip, is a miniaturized electronic circuit containing numerous transistors, resistors, capacitors, and other electronic components, all interconnected and fabricated on a single piece of semiconductor material, typically silicon. Essentially, it's a complete electronic circuit condensed onto a tiny piece of material, forming the foundation of modern electronics.

Here's a more detailed breakdown of their applications:

1. Consumer Electronics:
2. Automotive Industry:
3. Industrial Automation:

4. Other Applications: Lighting: ICs are used in LED lighting systems to control brightness and color. Smart Home Devices: ICs are used in thermostats, light switches, security systems, and other smart home devices. Scientific Instruments: ICs are used in various scientific instruments for data acquisition, analysis, and control.

In essence, integrated circuits have become the building blocks of modern electronics, enabling a vast array of technologies that impact almost every aspect of our lives.

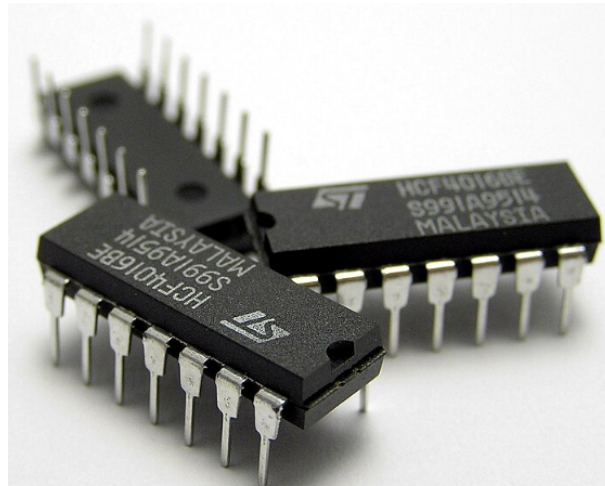


Figure 1.3.5: Dual in line package

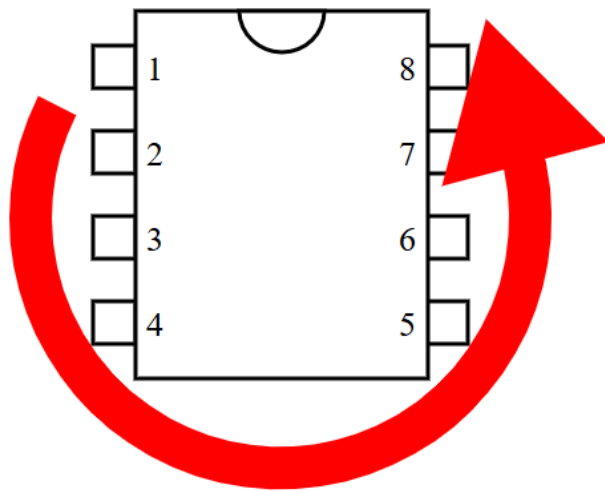


Figure 1.3.6: Pin numbering of an IC

1.3.5 Breadboard

A breadboard is a solder-less device used to build and test electronic circuits. It allows for temporary connections of components without the need for soldering, making it ideal for prototyping and experimenting with circuits.

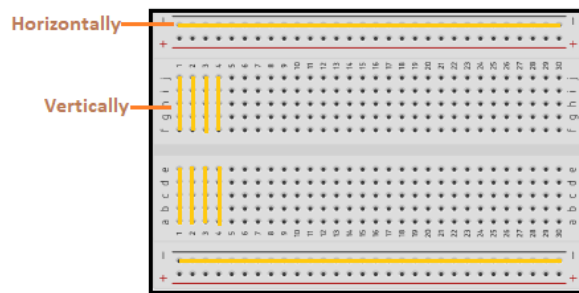


Figure 1.3.7: Breadboard

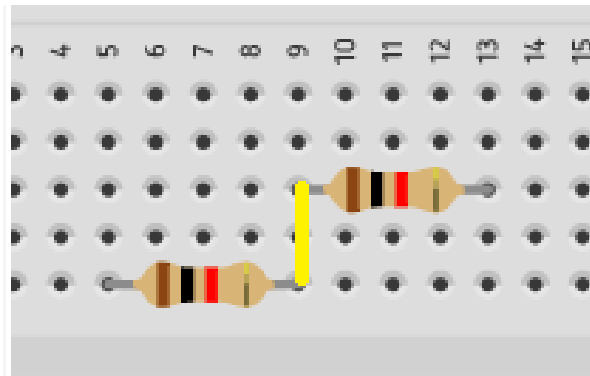


Figure 1.3.8: Connecting two resistors in series

1.3.6 Multimeter

A multimeter is an electronic measuring instrument that combines several functions in one unit, typically measuring voltage, current, and resistance. It is also known as a volt-ohm-milliammeter (VOM). Multimeters are used for troubleshooting and maintaining electrical systems and circuits in various applications.

Key Functions:

- Voltage Measurement:** Multimeters measure the electrical potential difference between two points in a circuit.
- Current Measurement:** They measure the flow of electrical charge through a circuit.
- Resistance Measurement:** Multimeters measure the opposition to the flow of current in a circuit.
- Other Measurements:** Some multimeters can also measure capacitance, frequency, and temperature.



Figure 1.3.9: Multimeter

1.3.7 DC Regulated power supply

DC power supply flows electric charge in one direction, so it supplies energy with fixed polarity. This power supply can obtain power from an AC or DC source. The battery is a very common example of a DC power supply.

A regulated power supply (also known usually as a linear power supply) ensures that the output current remains constant, even if the input changes, by converting unregulated AC (alternating current) to a constant DC (direct current).

1.4 Conclusion:



Figure 1.3.10: Power Supply

Experiment Number 2

Introduction to Function Generator and Digital Storage Oscilloscope(DSO)

2.1 Aim: To observe and measure the properties of different types of periodic signals using function generator and Oscilloscope

2.2 Equipment/Components required

Function Generator, Digital Storage Oscilloscope, Resistors, Probes, Breadboard, Connecting wires

2.3 Introduction

In order to perform an experiment, it is required to use different elements/components such as resistor, capacitor, breadboard, power supply and so on. This experiment presents the basic devices that are used in this lab to perform certain experiments.

2.4 Function Generator

A Function Generator is a scientific device that can generate a variety of different waveforms with a wide range of frequencies.

Waveforms:

Sine wave: This is generated from triangular wave by adding a pair of back to back diodes. Some specifications of this waveform are higher distortion as compared to sine waves produced by other test instruments

Triangular wave: The line obtained in this signal will not particularly be a straight line. This means there will be a departure from a straight line. If levels are around 90% of waveform amplitude then 99% linearity can be achieved.

Square wave: It categorizes one important specification which is edge rise and fall time which is significant in logic chips. Synchronous chips needing clock require an edge of specific



Figure 2.4.1: Function Generator

speed. A function generator can produce rise time and fall time of 100ns between 10 and 90% of the waveform.

Output symmetry(Duty Cycle) : Output symmetry is another important specification. Function generator provides a range over which we can control the symmetry of our output waveform. The average range is 20% - 80% with positive or negative 10% error.

DC offset: This specification is provided by some Function Generators. This allows to control the base voltage level of signal over a given range. The sample range can be around +5V or -5V.

2.5 Digital Storage Oscilloscope

A digital storage oscilloscope (DSO) is an electronic instrument that measures and records electrical signals. It converts the analog signal into a digital format and stores it in its digital memory, allowing for easy recall and analysis.

2.6 Activity to perform

1. Measure the voltage across each resistor in a series circuit consisting of 10 V DC source connected in series with two resistors with values $1.1\text{ k}\Omega$ and $2.7\text{ k}\Omega$.
2. Generate a sinusoidal signal of frequency 1 KHz with an amplitude of 5 V(p-p) and observe it on the DSO. Use two resistors of different values to monitor the voltage drop across each resistor on two channels.

2.7 Measurements/Observations

2.7.1 Measurements for Activity 1:

Supply Voltage V_s :

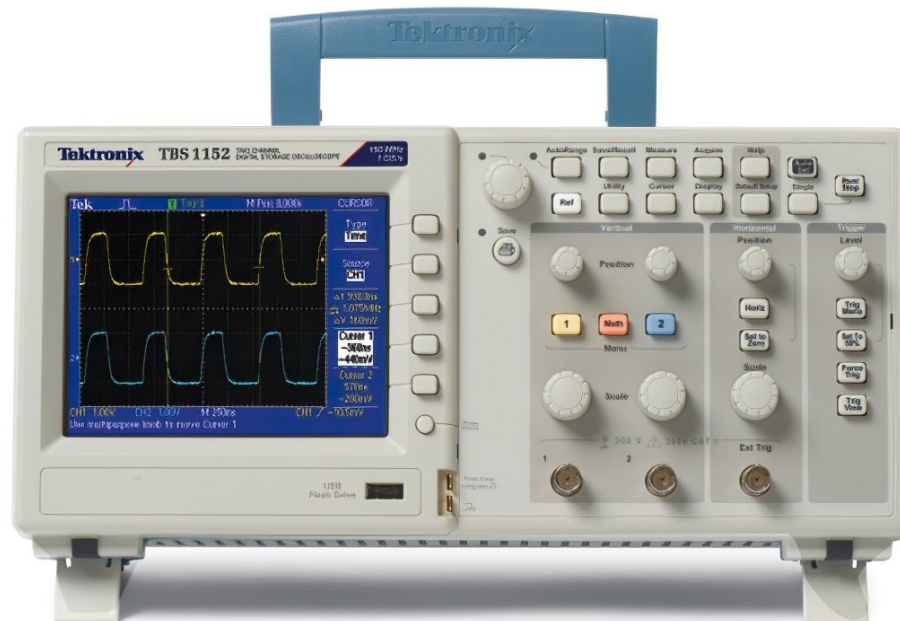


Figure 2.5.1: Digital Storage Oscilloscope

Resistor values: $R_1 =$ $R_2 =$ Voltage Across R_1 :Voltage Across R_2 :

2.7.2 Measurements for Activity 2:

Peak to peak Supply Voltage V_s :

Frequency:

Resistor values: $R_1 =$ $R_2 =$ Peak to peak Voltage Across R_1 :Peak to peak Voltage Across R_2 :

2.8 Conclusion:

Experiment Number 3

Verification of Basic Laws of Electricity

3.1 Aim: To verify Ohm's Law, KCL and KVL

3.2 Equipment/Components required

Regulated D.C Power Supply, Multimeter, Resistors, Connecting wires, Breadboard

3.3 Theory

Ohm's Law: Ohm's law states that the voltage across conducting materials is directly proportional to the current flowing through the material.

$$v = i \times R \quad (3.1)$$

Kirchhoff's Current Law:

The algebraic sum of the currents entering any node is zero.

$$\sum_{n=1}^{n=N} i_n = 0 \quad (3.2)$$

Kirchhoff's Voltage Law:

The algebraic sum of the voltages around any closed path is zero.

$$\sum_{n=1}^{n=N} v_n = 0 \quad (3.3)$$

3.4 Activity 1: Verification of Ohm's Law

3.4.1 Procedure

1. Connect a variable voltage source across two series resistors as shown in Fig. 3.4.1 .
2. Make sure that an ammeter(multimeter to measure current) is connected in series with the resistors to measure the current passing through it.

18 EXPERIMENT NUMBER 3. VERIFICATION OF BASIC LAWS OF ELECTRICITY

3. Make sure that a voltmeter is connected in parallel (Use DSO Channel 1) with the 470Ω resistor to measure the voltage drop across it.
4. Gradually change the supply voltage V_s from 0 to 15 V in step of 2 Volt and measure the voltage drop across the resistor and current passing through it. Note down the readings in Tab 3.1

S.No	Supply Voltage V_s (V)	Current through 470Ω (mA)	Voltage Across 470Ω (V)
1	0		
2	2		

Table 3.1: v versus i

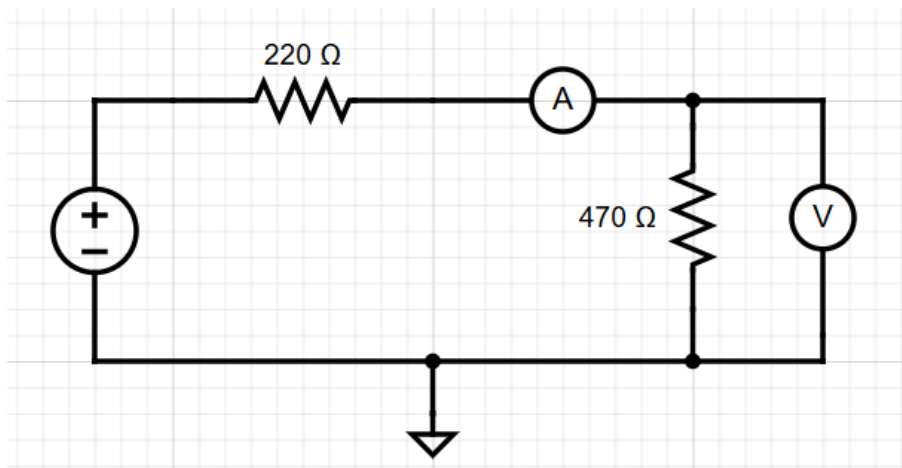


Figure 3.4.1: Circuit for Ohm's Law

3.4.2 Calculations

1. Plot the graph between v and i on a graph sheet.
2. Determine the slope of the line.
3. Compare this value with that of the measured resistance using multimeter/color coding.

3.4.3 Conclusion:

3.5 Activity 2: Verification of KCL

3.5.1 Procedure:

1. Connect the circuit as shown in Fig. 3.6.1 .

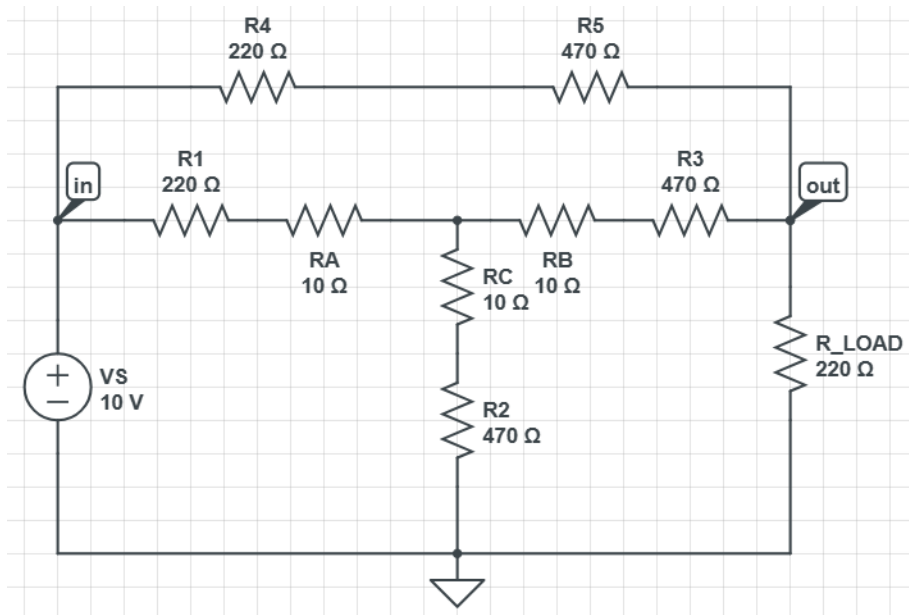


Figure 3.5.1: Circuit for KCL verification

2. Use multimeter to measure the voltage drop across R_A , R_B and R_C .
3. Divide the voltage measure in previous step to measure currents i_{R_A} , i_{R_B} and i_{R_C} . Note the readings as per Tab. 3.2.

S.No	Measured Voltage V_{R_x} (V)	Current through R_A (mA))
1	$V_{RA} =$	I_{RA}
2	$V_{RB} =$	I_{RB}
3	$V_{RC} =$	I_{RC}

Table 3.2: Currents through different resistors

3.5.2 Calculations:

Verify the equation:

$$\sum_{n=1}^{n=N} i_n = 0 \quad (3.4)$$

3.6 Activity:3 Verification of KVL

3.6.1 Procedure:

1. Connect the circuit shown in Fig. 3.6.1 .

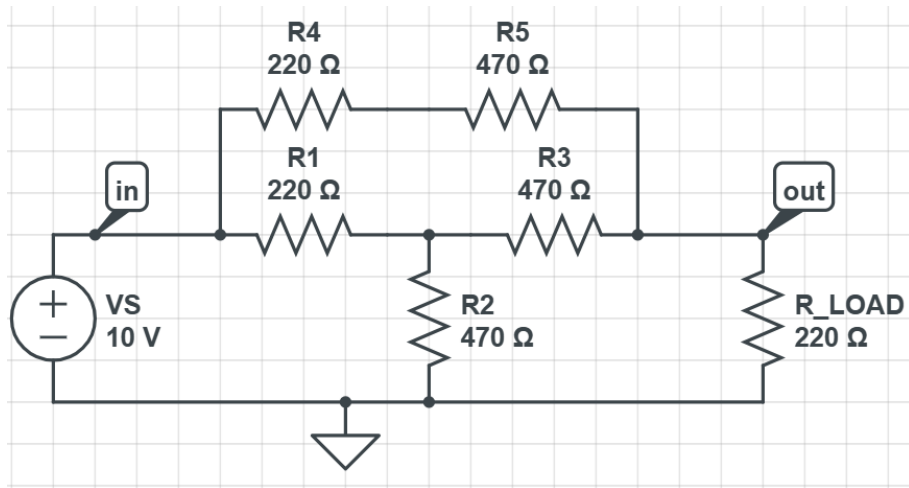


Figure 3.6.1: Circuit for KVL verification

2. Measure the voltage across each resistor carefully paying attention to polarity. Note the readings as per Tab. 3.3.

S.No	Measured Voltage V_{Rx} (V)	Measured Voltage V_{Rx} (V)
1	$V_{R1} =$	$V_{R2} =$
2	$V_{R3} =$	$V_{R4} =$
3	$V_{R5} =$	$V_{RLOAD} =$

Table 3.3: Voltages across different resistors

3.6.2 Calculations:

Verify the equation for two different loops.

$$\sum_{n=1}^{n=N} v_n = 0 \quad (3.5)$$

3.6.3 Conclusion:

Experiment Number 4

Verification of the superposition principle

4.1 Aim: To verify the superposition principle

4.2 Equipment/Components required

Regulated D.C Power Supply, Multimeter, Resistors, Connecting wires, Breadboard

4.3 Theory

In any linear resistive network, the voltage across or the current through any resistor or source may be calculated by adding algebraically all the individual voltages or currents caused by the separate independent sources acting alone, with all other independent voltage sources replaced by short circuits and all other independent current sources replaced by open circuits.

A voltage source set to zero acts like a short circuit.

$$v = 0 \quad \text{AND} \quad i \neq 0 \quad \implies R = 0 \quad (4.1)$$

A current source set to zero acts like an open circuit.

$$v \neq 0 \quad \text{AND} \quad i = 0 \quad \implies R \rightarrow \infty \quad (4.2)$$

4.4 Activity 1 : Verification of Superposition principle

4.4.1 Procedure

Case 1: Both Sources are activated

1. Connect the circuit as shown in Fig. 4.4.1.
2. Make sure that both the voltage sources are activated simultaneously with proper polarity.
3. Measure the voltage drop v_x using the multimeter as a voltmeter. Only one reading for v_x .

22 EXPERIMENT NUMBER 4. VERIFICATION OF SUPERPOSITION PRINCIPLE

Note down the readings as shown in Table. 4.1

S.No	Voltage (V)	v_x
1	$V_{s1}=12$	
2	$V_{s2} = +15$	

Table 4.1: Both Sources are activated

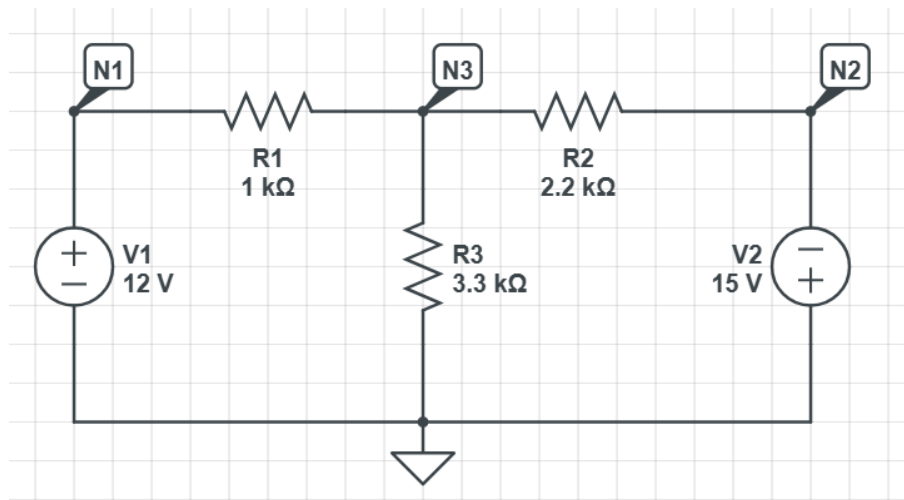


Figure 4.4.1: Both Sources are activated

Case 2: Only +12 V source is activated

1. Connect the circuit as shown in Fig. 4.4.2.
2. Make sure that +12 V voltage source is activated while +15 V voltage source has been deactivated.
3. Measure the voltage drop v'_x using the multimeter as a voltmeter. Only one reading for v'_x .

Note down the readings as shown in Table. 4.2

S.No	Voltage (V)	v'_x
1	$V_{s1} = 12$	
2	$V_{s2} = 0$	

Table 4.2: +12 V source is activated

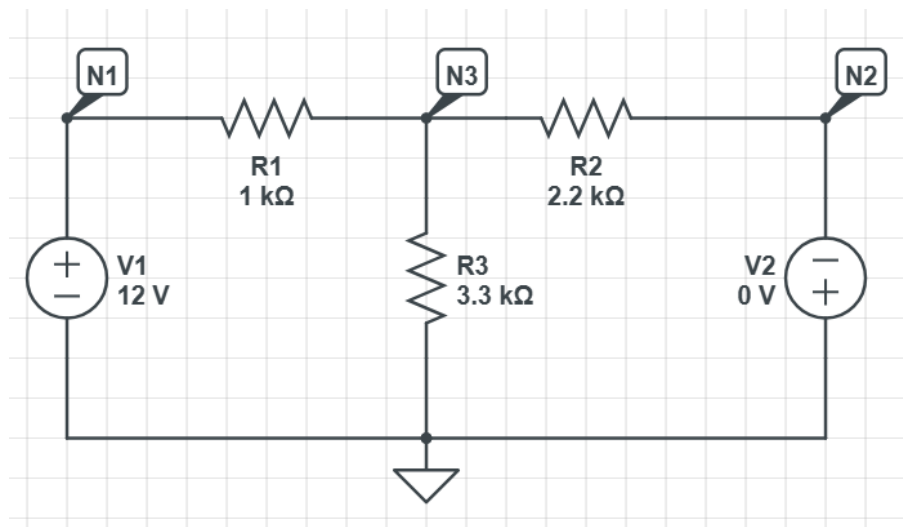


Figure 4.4.2: +12 V source is activated

Case 3: Only +15 V source is activated

1. Connect the circuit as shown in Fig. 4.4.3.
2. Make sure that +15 V voltage source is activated while +12 V voltage source has been deactivated.
3. Measure the voltage drop v_x'' using the multimeter as a voltmeter. Only one reading for v_x'' .

Note down the readings as shown in Table. 4.3

S.No	Voltage (V)	v_x''
1	$V_{s1}=0$	
2	$V_{s2} = +15$	

Table 4.3: +12 V source is activated

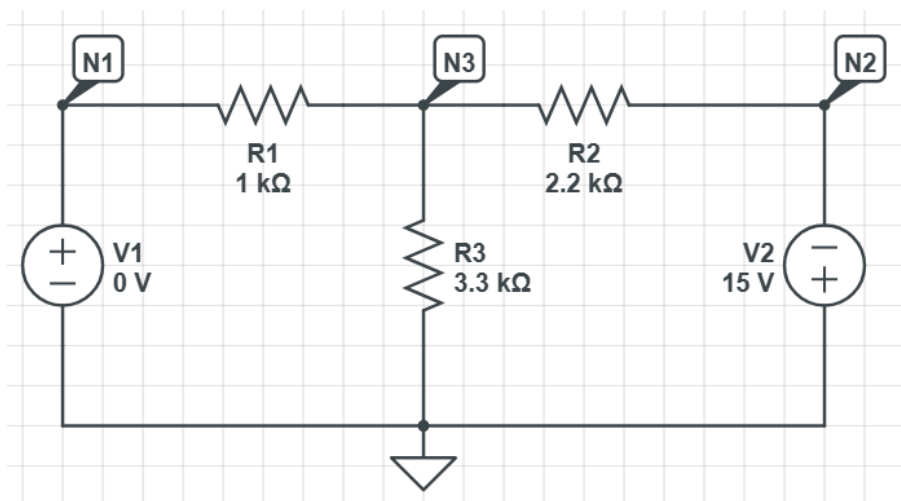


Figure 4.4.3: +15 V source is activated

4.4.2 Calculations

1. Add v_x' with v_x'' and verify that this is close to v_x
2. Use the above values to prove that superposition principle cannot be applied to power calculations.
 - * Calculate the powers dissipated across 3.3 k Ω from case 1, P_1 , case 2, P_2 , and case 3, P_3 .
 - * Observe whether , $P_1 = P_2 + P_3$.

4.5 Activity 2: Find the value of v_1 such that the current passing through $3.3\text{ K}\Omega$ is zero

4.5.1 Procedure

1. Connect the circuit as shown in Fig. 4.4.3.
2. Gradually change the voltage such that the potential difference between $N3$ and Ground node is zero.

4.5.2 Conclusion:

Experiment Number 5

Thevenin's Theorem

5.1 Aim: To obtain the Thevenin's equivalent circuit of a given circuit and verify the maximum power transfer theorem

5.2 Equipment/Components required

Regulated D.C Power Supply, Multimeter, Resistors, Connecting wires, Breadboard

5.3 Theory

5.3.1 Thevenin's Equivalent

A Statement of Thevenin's Theorem

1. Given any linear circuit, rearrange it in the form of two networks, A and B, connected by two wires.

Network A is the network **to be simplified**; B will be left untouched.

2. Disconnect network B. Define a voltage v_{oc} as the voltage now appearing across the terminals of network A.

3. Turn off or "**zero out**" every **independent source** in network A to form an inactive network. **Leave dependent sources unchanged.**

4. Connect an independent voltage source with value V_{oc} in series with the inactive network.

Do not complete the circuit; leave the two terminals disconnected.

5. Connect network B to the terminals of the new network A.

All currents and voltages in B will remain unchanged.

5.4 Activity : To determine V_{oc} and R_{TH}

5.4.1 Procedure

1. Connect the circuit as shown in Fig. 5.4.1. Measure the voltage drop across the R_L . Designate this as V_{R_L} .
2. To measure the V_{oc} , remove the R_L and note down the voltage across the open circuited terminals. This is V_{oc} .
3. To find R_{TH} , deactivate the voltage source(i.e., replace with a short circuit) and measure the resistance between the open circuited terminals using a multimeter. This is R_{TH} .
4. Connect the Thevenin's equivalent circuit with V_{TH} and R_{TH} along with R_L . Measure the voltage drop across the R_L . This is the load voltage with Thevenin's equivalent. It should be close to V_{R_L} .

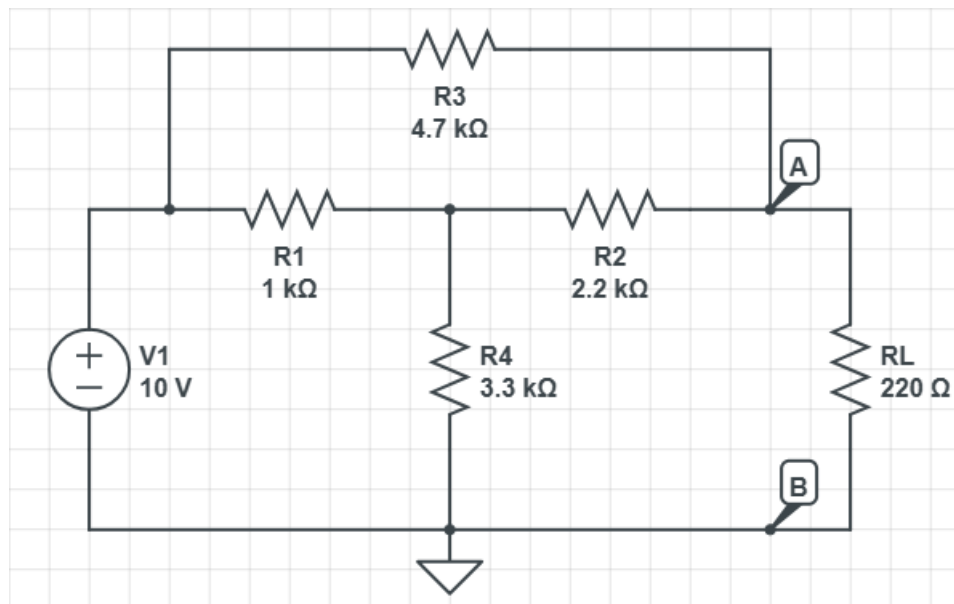


Figure 5.4.1: Circuit diagram

5.5 Activity : To determine R_{TH} by finding V_{oc}/I_{sc}

5.5.1 Procedure

1. Find v_{oc} as mentioned earlier.
2. Replace the R_L with a short circuit and measure the current passing through it. In order to do so, replace the R_L with an ammeter and measure the current passing through it.
3. Calculate $R_{TH} = v_{oc}/i_{sc}$
4. Compare this with that of earlier equivalent resistance.

5.6 Activity : To determine R_{TH} by finding V_{test}/I_{test}

5.6.1 Procedure

1. Deactivate the voltage source and connect a test voltage source of 1 V in place of R_L .
2. Determine the current supplied by the test voltage source.
3. Calculate $R_{TH} = v_{test}/i_{test}$
4. Compare this with that of earlier equivalent resistance.

5.7 Activity : To determine v_{R_L} from Thevenin's equivalent circuit

5.7.1 Procedure

1. Connect the equivalent circuit with v_{TH} , R_{TH} and R_L all in series.
2. Measure the voltage drop across the R_L .
3. Compare this with that of earlier voltage drop across R_L

5.7.2 Conclusion:

Experiment Number 6

PN-junction diode characteristics

6.1 Aim: To study the forward bias I-V (current-voltage) characteristics of a simple p-n junction diode. Also to determine the forward resistance of the diode

6.2 Equipment/Components required

Variable DC Power Supply (0-32V), Digital Multimeter as Voltmeter and Ammeter, Diode IN4007, Resistor 1K, Ammeter (0-100mA)

6.3 Theory

A P-N junction diode is a semiconductor device formed by joining a p-type and an n-type semiconductor material. It acts as a one-way valve for electrical current, allowing it to flow easily in one direction (forward bias) while significantly restricting flow in the opposite direction (reverse bias). This behaviour is due to the unique properties of the depletion region formed at the junction.

6.3.1 V-I Characteristics

The diode's behaviour is non-linear Fig. 6.3.1. It doesn't conduct until the forward voltage reaches a certain threshold (e.g., about 0.7V for silicon). Once the threshold is reached, the current increases rapidly with increasing forward voltage. In reverse bias, a small reverse saturation current flows initially, but if the reverse voltage becomes too high (breakdown voltage), the diode can be damaged.

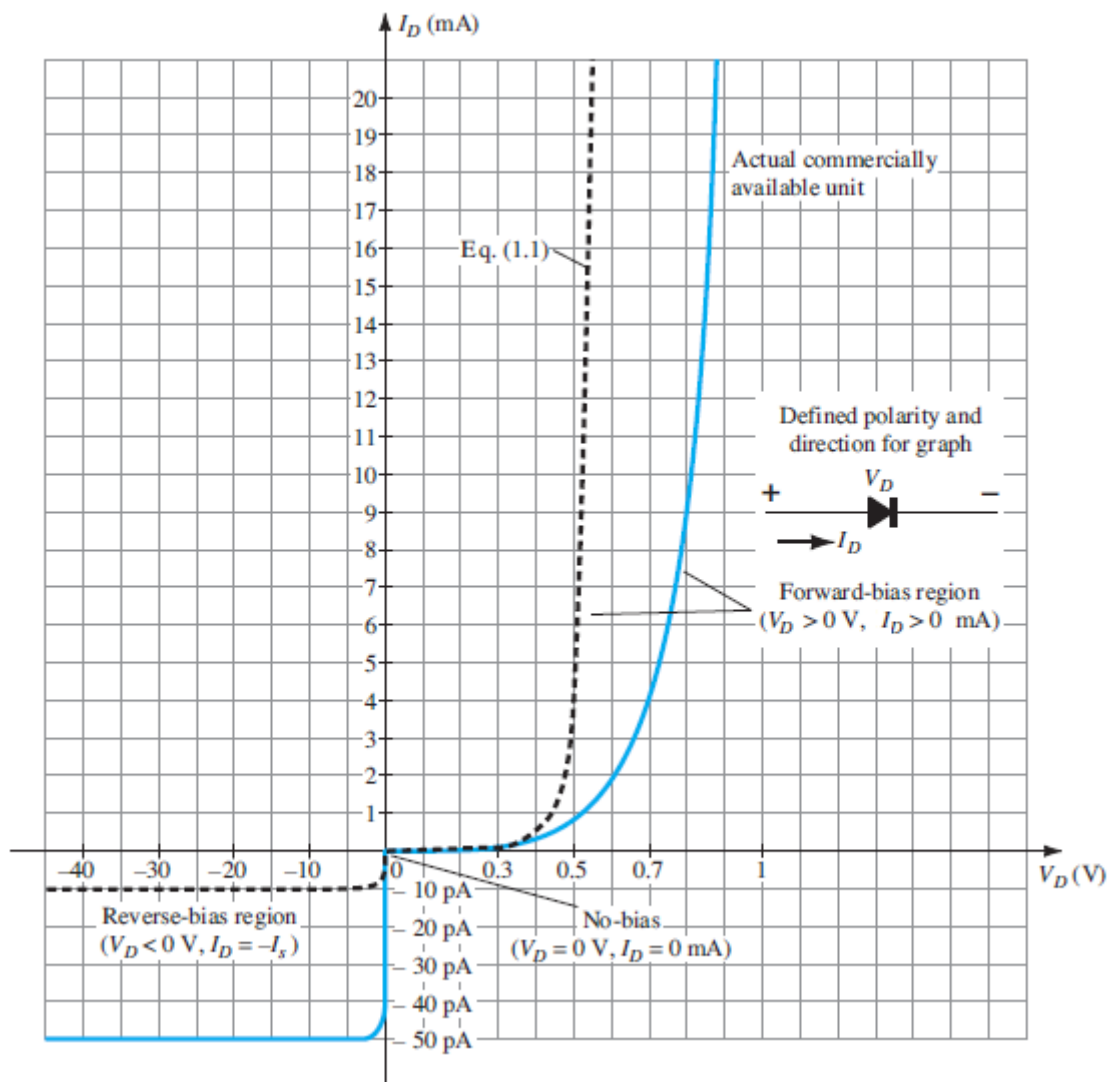


Figure 6.3.1: Silicon semiconductor diode characteristics.

6.4 Activity 1 : Forward bias Characteristics of P-N junction diode.

6.4.1 Procedure

1. Connect the circuit as shown in Fig. 6.4.1.
2. Vary the input voltage V_s in small steps from 0 to 2V.
3. Note down the readings V_d and I_d .
4. Note down the cut-in voltage at which the diode starts conducting.
5. Plot the graph (V_d I_d)

Note down the readings as shown in Table. 10.1

S.No	v_s	v_d	i_d
1			
2			

Table 6.1: Observed Readings

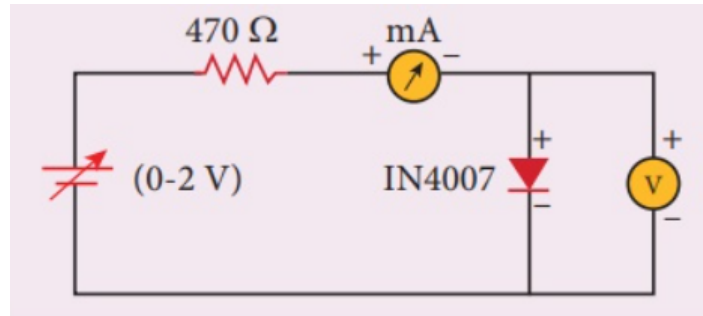


Figure 6.4.1: Circuit diagram

6.5 Activity 2 : Reverse bias Characteristics of P-N junction diode.

6.5.1 Procedure

1. Interchange the polarity of the dc supply by connecting -ve polarity to p-side and +ve polarity to n-side of the pn-junction diode.
2. Increase the input voltage V_s stepwise in reverse bias.
3. Note down the readings V_d and I_d .
4. Note down the break down voltage at which the diode starts conducting
5. Plot the graph (V_d and I_d)

Note down the readings as shown in Table. 10.1

S.No	v_s	v_d	i_d
1			
2			

Table 6.2: Observed Readings

6.5.2 Resistance Calculations

DC or Static Resistance

$$R_D = \frac{V_D}{I_D} \quad (6.1)$$

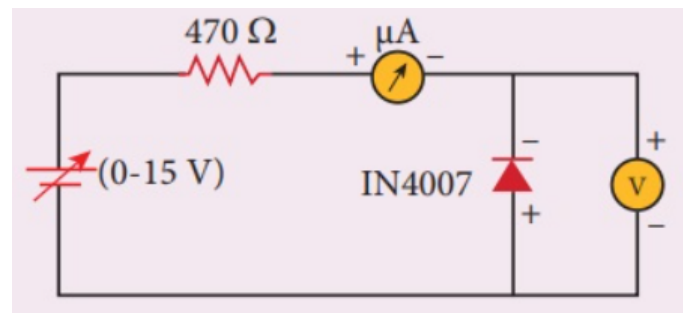


Figure 6.5.1: Circuit diagram

AC or Dynamic Resistance

$$r_d = \frac{\Delta V_d}{\Delta I_d} \quad (6.2)$$

Different resistance levels are indicated in Fig. 6.5.2

Type	Equation	Special Characteristics	Graphical Determination
DC or static	$R_D = \frac{V_D}{I_D}$	Defined as a point on the characteristics	
AC or dynamic	$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26 \text{ mV}}{I_D}$	Defined by a tangent line at the Q-point	
Average ac	$r_{av} = \left. \frac{\Delta V_d}{\Delta I_d} \right _{\text{pt. to pt.}}$	Defined by a straight line between limits of operation	

Figure 6.5.2: Resistance Levels

6.5.3 Conclusion:

Experiment Number 7

Rectifier circuits

7.1 Aim: To implement a Half-Wave and Full-wave rectifier circuits.

7.2 Equipment/Components required

Variable DC Power Supply (0-32V), Digital Multimeter as Voltmeter and Ammeter, Diode IN4007, Resistor 1K, Ammeter (0-100mA)

7.3 Theory

A half-wave rectifier converts an AC signal to DC by passing either the negative or positive half-cycle of the waveform and blocking the other. Half-wave rectifiers can be easily constructed using only one diode, but are less efficient than full-wave rectifiers.

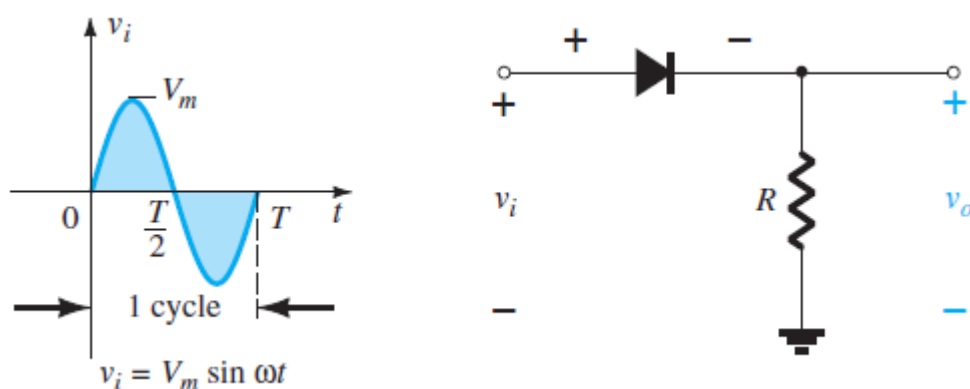


Figure 7.3.1: Half-wave rectifier.

Alternating current (AC) periodically changes direction, and a rectifier converts this signal to a direct current (DC), which only flows in one direction. A half-wave rectifier does this by removing half of the signal. A full-wave rectifier converts the full input waveform to one of constant polarity by reversing the direction of current flow in one half-cycle.

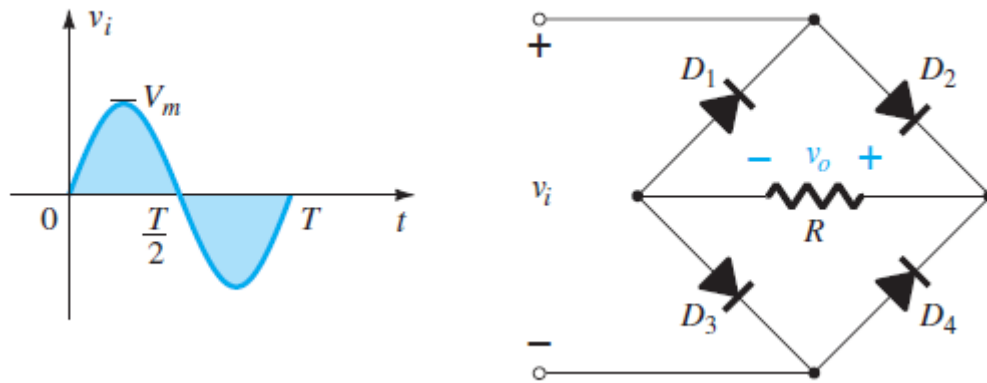
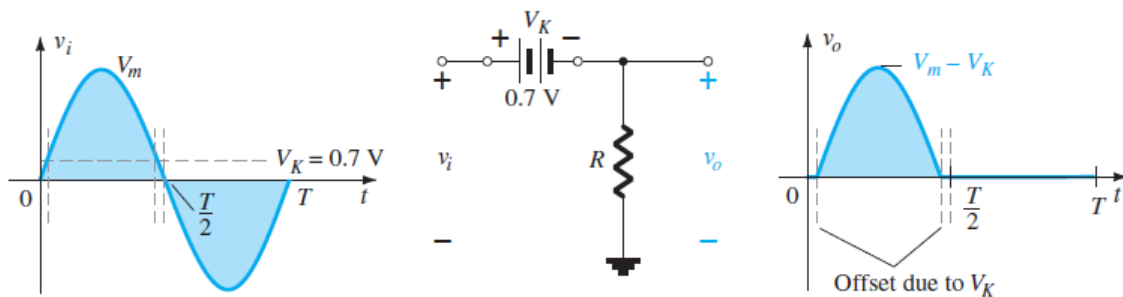


Figure 7.3.2: Full-wave rectifier.

7.4 Activity 1 : Half-wave rectifier circuit

7.4.1 Procedure

1. Connect the circuit as shown in Fig. 7.3.1 with certain modifications.
2. Connect Ch-1 of DSO to input and Ch-2 of DSO to output voltages.
3. Observe the waveforms carefully by aligning both input and output at the middle of the display.
4. Note down Offset due to V_K from the waveforms.

Figure 7.4.1: Effect of V_K on half-wave rectified signal.

7.5 Activity 2 : Full-wave rectifier circuit

7.5.1 Procedure

1. Connect the circuit as shown in Fig. 7.3.2 with certain modifications.
2. Connect Ch-1 of DSO to input and Ch-2 of DSO to output voltages.

3. Observe the waveforms carefully by aligning both input and output at the middle of the display.
4. Note down Offset due to V_K from the waveforms.

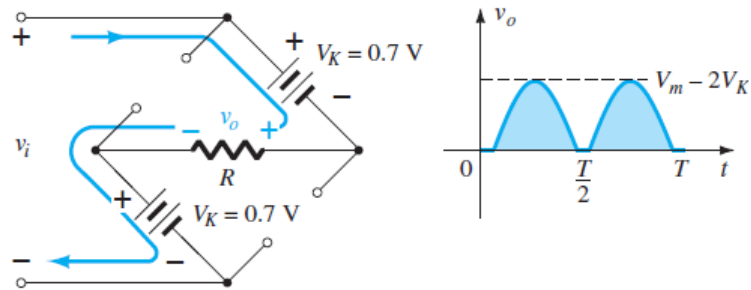


Figure 7.5.1: Effect of V_K on full-wave rectified signal.

7.5.2 Conclusion:

Experiment Number 8

Zener diode characteristics and application

8.1 Aim: To study the reverse bias I-V (current-voltage) characteristics of a Zener diode. Also determine the breakdown voltage, static and dynamic resistances.

8.2 Equipment/Components required

Variable DC Power Supply (0-32V), Digital Multimeter as Voltmeter and Ammeter, Diode IN4007, Resistor 1K, Ammeter (0-100mA)

8.3 Theory

A Zener diode is a specialized diode designed to operate in reverse breakdown, maintaining a relatively stable voltage across it even as current varies significantly. This characteristic makes it ideal for voltage regulation applications. Key characteristics include a defined breakdown voltage (V_z), power rating, temperature stability, and ability to operate in both forward and reverse bias conditions.

8.3.1 V-I Characteristics

1. Zener Breakdown Voltage (V_z): This is the specific reverse voltage at which the diode begins to conduct significantly, offering a stable voltage output. Zener voltages can range from a few volts to several hundred, and are crucial for voltage regulation.

2. Forward Bias: Like a standard diode, a Zener diode conducts when forward biased, with a voltage drop (typically 0.7V for silicon).

3. Reverse Bias: In reverse bias, a small leakage current flows until the voltage reaches the Zener voltage (V_z). Once V_z is reached, the diode enters breakdown and conducts, maintaining a relatively stable voltage across itself.

4. Power Rating: This indicates the maximum power the diode can dissipate without damage, calculated as the product of voltage and current.

5. Temperature Stability: Zener diodes, especially those around 5.5V, exhibit good temperature stability due to opposing temperature coefficients of the Zener and avalanche effects.

6. Voltage Tolerance: Specifies the acceptable variation in the Zener voltage, typically around $\pm 5\%$.

7. Zener Resistance (R_Z): This is the dynamic resistance of the diode in the breakdown region, affecting the output voltage stability.

8. Current Handling Capacity: Zener diodes have a minimum current (I_{Zmin}) required for breakdown and a maximum current (I_{Zmax}) they can handle.

Applications: Voltage Regulation: Zener diodes are widely used to maintain a stable voltage across a load, even with fluctuating input voltage or current.

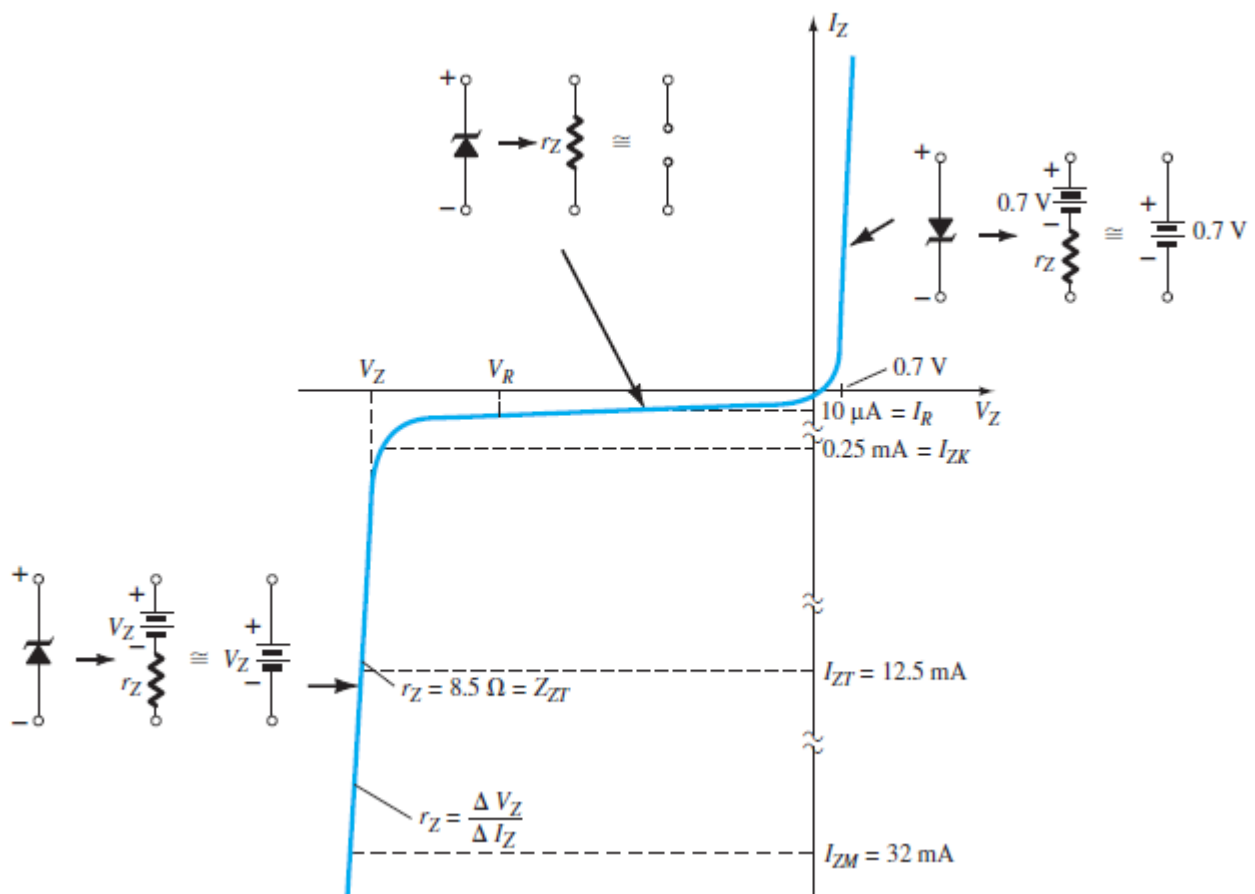


Figure 8.3.1: Zener diode characteristics with the equivalent model for each region.

8.4 Activity 1 : Forward bias Characteristics of Zener junction diode.

8.4.1 Procedure

1. Connect the circuit as shown in Fig. 8.4.1.

2. Vary the input voltage V_s in small steps from 0 to 2V.
3. Note down the readings V_d and I_d .
4. Note down the cut-in voltage at which the diode starts conducting.
5. Plot the graph (V_d I_d)

Note down the readings as shown in Table. 10.1

S.No	v_s	v_d	i_d
1			
2			

Table 8.1: Observed Readings

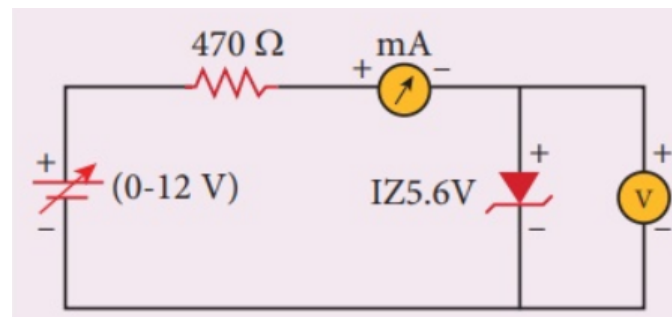


Figure 8.4.1: Circuit diagram

8.5 Activity 2 : Reverse bias Characteristics of P-N junction diode.

8.5.1 Procedure

1. Interchange the polarity of the dc supply by connecting -ve polarity to p-side and +ve polarity to n-side of the pn-junction diode.
2. Increase the input voltage V_s stepwise in reverse bias.
3. Note down the readings V_d and I_d .
4. Note down the break down voltage at which the diode starts conducting
5. Plot the graph (V_d and I_d)

Note down the readings as shown in Table. 10.1

S.No	v_s	v_d	i_d
1			
2			

Table 8.2: Observed Readings

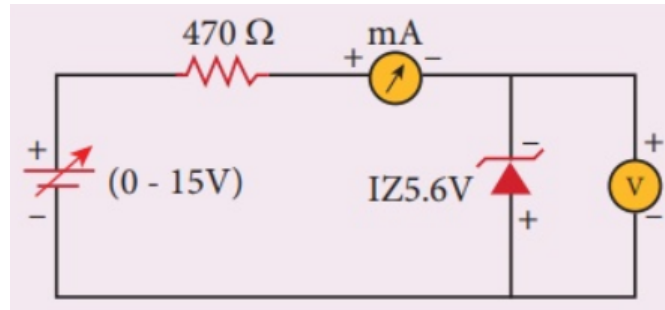


Figure 8.5.1: Circuit diagram

8.5.2 Resistance Calculations

DC or Static Resistance

$$R_Z = \frac{V_Z}{I_Z} \quad (8.1)$$

AC or Dynamic Resistance

$$r_Z = \frac{\Delta V_Z}{\Delta I_Z} \quad (8.2)$$

8.6 Activity 3: Zener Diode as a voltage regulator

Analyse the voltage regulator for variable load resistance.

8.6.1 Procedure

1. Connect the circuit as shown in Fig. 8.6.1
2. Change the value of load resistance from low values to higher values
3. Note down the readings of V_{R_L} and I_Z .
4. Plot the graph between R_L and V_{R_L}

S.No	R_L	V_L	I_L
1			
2			

Table 8.3: Observed Readings

8.6.2 Conclusion:

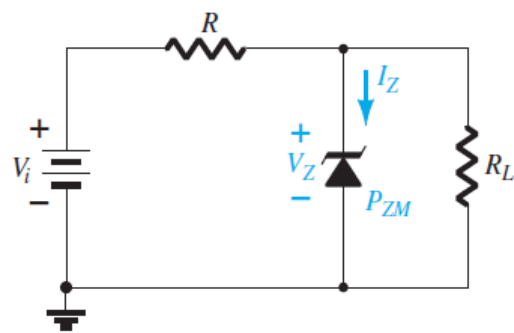


Figure 8.6.1: Circuit diagram

Experiment Number 9

Clipper and Clamper Circuits

9.1 Aim: To implement diode clipper and clampers circuits and observe the output waveforms on DSO

9.2 Equipment/Components required

Variable DC Power Supply (0-32V), Digital Multimeter as Voltmeter and Ammeter, Diode IN4007, Resistor 1K, Ammeter (0-100mA)

9.3 Theory

Clippers are networks that employ diodes to "clip" away a portion of an input signal without distorting the remaining part of the applied waveform.

Clamping networks have a capacitor connected directly from input to output with a resistive element in parallel with the output signal. The diode is also in parallel with the output signal but may or may not have a series dc supply as an added element.

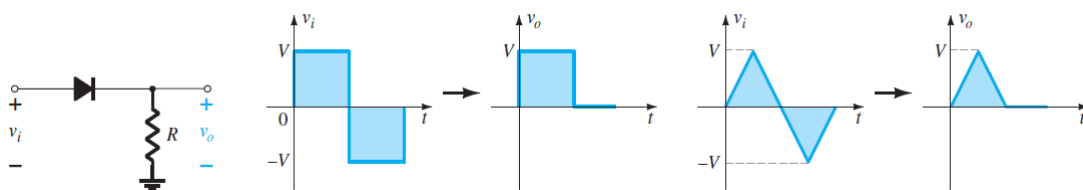


Figure 9.3.1: Series clipper.

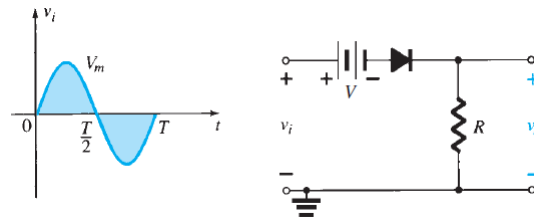


Figure 9.3.2: Series clipper with a dc supply.

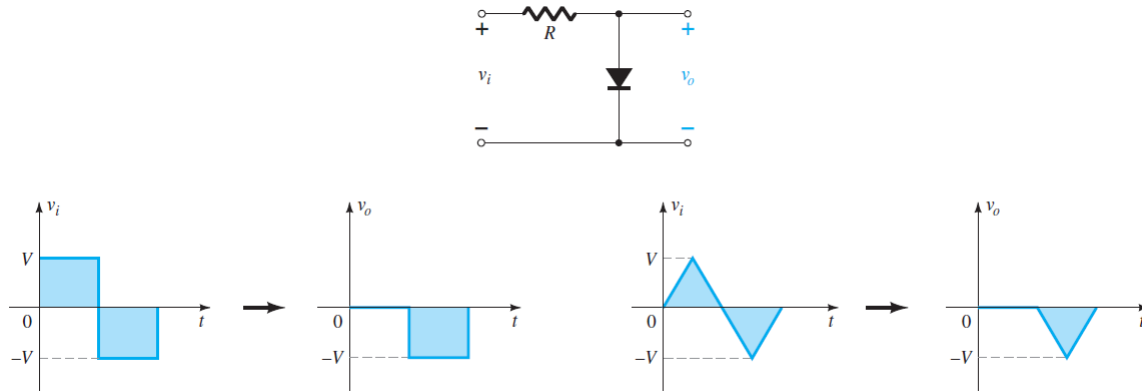


Figure 9.3.3: Response to a parallel clipper.

9.4 Activity 1 : To determine the output voltage v_o for the network shown below.

9.4.1 Procedure

1. Connect the circuit as shown in Fig. 9.4.1 with certain modifications.
2. Connect Ch-1 of DSO to input and Ch-2 of DSO to output voltages.
3. Observe the waveforms carefully by aligning both input and output at the middle of the display.
4. Use tracing paper to record the waveforms on the DSO.

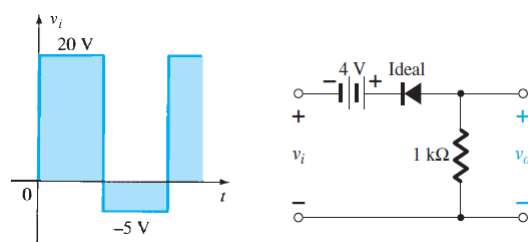


Figure 9.4.1: Response to a series clipper.

9.5 Activity 2 : To determine the output voltage v_o for the network shown below.

9.5.1 Procedure

1. Connect the circuit as shown in Fig. 9.5.1 with certain modifications.
2. Connect Ch-1 of DSO to input and Ch-2 of DSO to output voltages.
3. Observe the waveforms carefully by aligning both input and output at the middle of the display.
4. Use tracing paper to record the waveforms on the DSO.

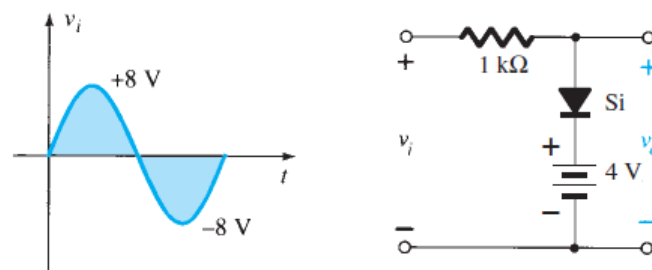


Figure 9.5.1: Response to a parallel clipper.

9.6 Activity 3 : To determine the output voltage v_o for the network shown below.

9.6.1 Procedure

1. Connect the circuit as shown in Fig. 9.6.1 with certain modifications.
2. Connect Ch-1 of DSO to input and Ch-2 of DSO to output voltages.
3. Observe the waveforms carefully by aligning both input and output at the middle of the display.
4. Use tracing paper to record the waveforms on the DSO.

9.7 Activity 4 : To determine the output voltage v_o for the network shown below.

9.7.1 Procedure

1. Connect the circuit as shown in Fig. 9.7.1 with certain modifications.

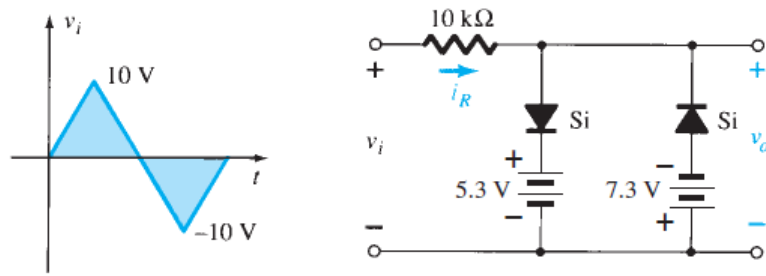


Figure 9.6.1: Response to a double sided clipper.

2. Connect Ch-1 of DSO to input and Ch-2 of DSO to output voltages.
3. Observe the waveforms carefully by aligning both input and output at the middle of the display.
4. Use tracing paper to record the waveforms on the DSO.

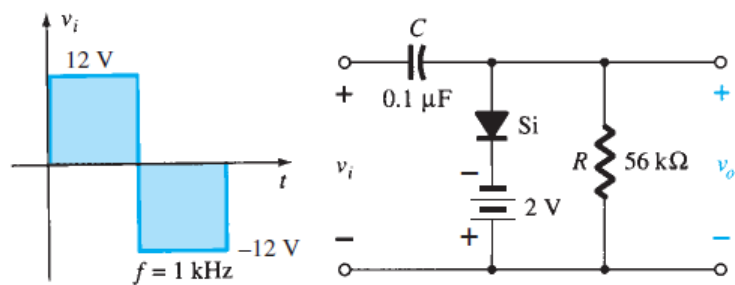


Figure 9.7.1: Clamper Circuit

9.7.2 Conclusion:

Experiment Number 10

Common Emitter Configuration

10.1 Aim: To plot input and output characteristics of a transistor in Common- Emitter configuration.

10.2 Equipment/Components required

Dual power supply, multimeter, Breadboard, NPN Transistor, Resistors, Probes and connecting wires as required.

10.3 Theory

The most frequently encountered transistor configuration appears in Fig. 10.3.1. It is called the common-emitter configuration because the emitter is common to both the input and output terminals (in this case common to both the base and collector terminals). Two sets of characteristics are again necessary to describe fully the behaviour of the common-emitter configuration: one for the input or base–emitter circuit and one for the output or collector–emitter circuit. These are represented in Fig. 10.3.2.

Clamping networks have a capacitor connected directly from input to output with a resistive element in parallel with the output signal. The diode is also in parallel with the output signal but may or may not have a series dc supply as an added element.

10.4 Activity 1 : To determine the input characteristic of npn transistor

10.4.1 Procedure

1. Connect the circuit as shown in Fig. 10.3.1 with certain modifications.
2. Make $V_{CE} = 2V$ with help of V_{CC} .
3. Now increase V_{BB} voltage in steps of 0.2V to 1V and then steps of 0.5V to 10V and then steps of 2V to 20V.

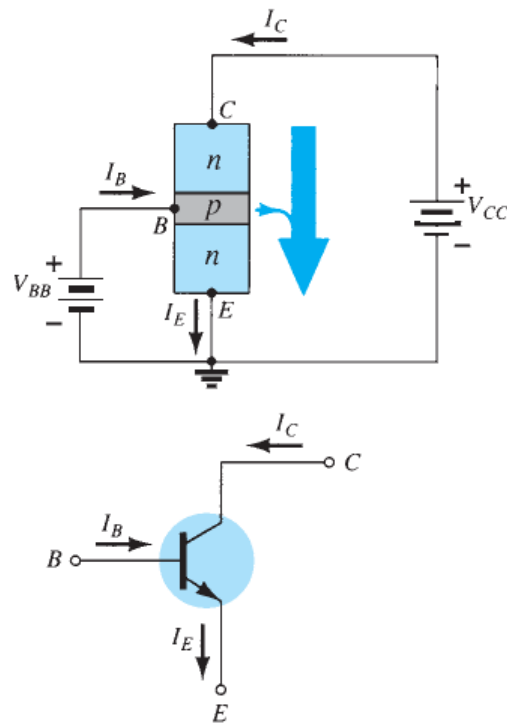


Figure 10.3.1: Notation and symbols used with the common-emitter configuration.

4. Measure V_{BE} and I_B and tabulate them.
5. Maintain V_{CE} constant with the help of V_{CC}
6. Plot the graph between V_{BE} and I_B
7. Now make $V_{CE} = 5V$ by varying V_{CC}
8. Repeat steps 3 to 6.

Note down the readings as shown in Table. 10.1

S.No	V_{BB}	V_{BE}	$I_B (\mu A)$
1			
2			

Table 10.1: Observed Readings for constant value of $V_{CE} = 10V$

10.5 Activity 2 : To determine the output characteristic of npn transistor

10.5.1 Procedure

1. Connect the circuit as shown in Fig. 10.3.1 with certain modifications.
2. Make $I_B = 20\mu A$ and maintain in constant using V_{BB} .

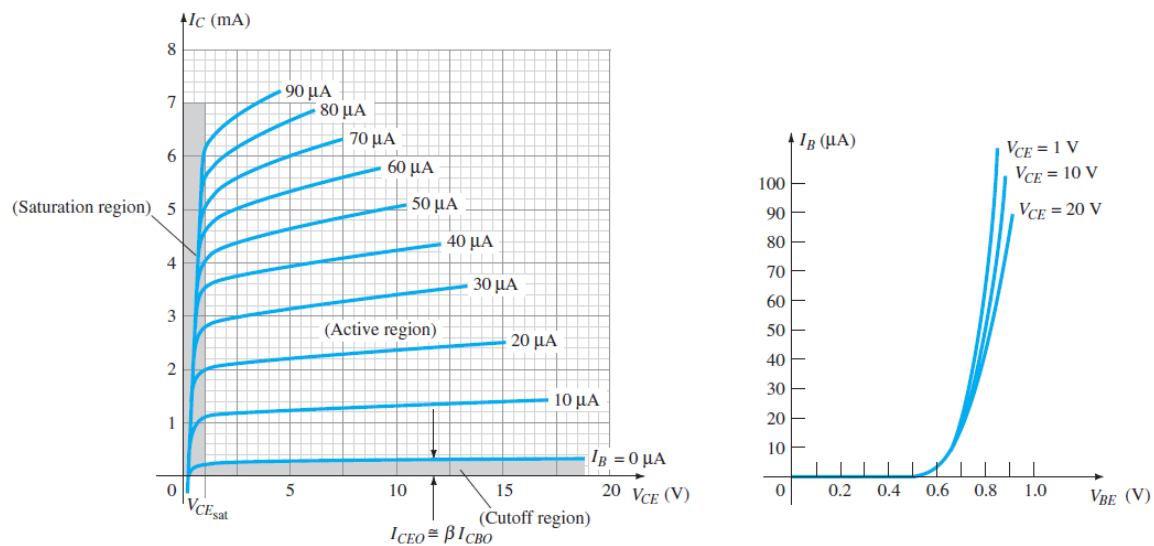


Figure 10.3.2: Collector characteristics and base characteristics.

3. Now vary V_{CC} in steps of 0.5V to 5V and then steps of 2V to 30V .
4. Measure I_C and V_{CE} keeping I_B constant and note down the readings and tabulate them.
5. Plot the graph between V_{CE} and I_C .
6. Now make $I_B = 40\mu A$ and repeat the steps from 3 to 5.

Note down the readings as shown in Table. 10.2

S.No	V_{CC}	V_{CE}	I_C (μA)
1			
2			

Table 10.2: Observed Readings for constant value of $I_B = 20\mu A$

10.5.2 Conclusion:

Experiment Number 11

Op-Amp amplifiers

11.1 Aim: To realize inverting and non-inverting amplifier configuration using Op-Amp IC $\mu A - 741$.

11.2 Equipment/Components required

Dual power supply, Function generator, DSO, Breadboard, IC -741, Resistors, Probes and connecting wires as required.

11.3 Theory

An operational amplifier, or op-amp, is a very high gain differential amplifier with high input impedance and low output impedance. Typical uses of the operational amplifier are to provide voltage amplitude changes (amplitude and polarity), to generate oscillations, filter signals, and serve as the basis of many types of instrumentation circuits. An op-amp contains a number of differential amplifier stages to achieve a very high voltage gain.

11.3.1 Inverting Amplifier

The output is obtained by multiplying the input by a fixed or constant gain, set by the input resistor (R_1) and feedback resistor (R_f) –this output also being inverted from the input. Fig. 11.3.1.

11.3.2 Non-Inverting Amplifier

The connection of Fig. 11.3.2 a shows an op-amp circuit that works as a non-inverting amplifier or constant-gain multiplier.

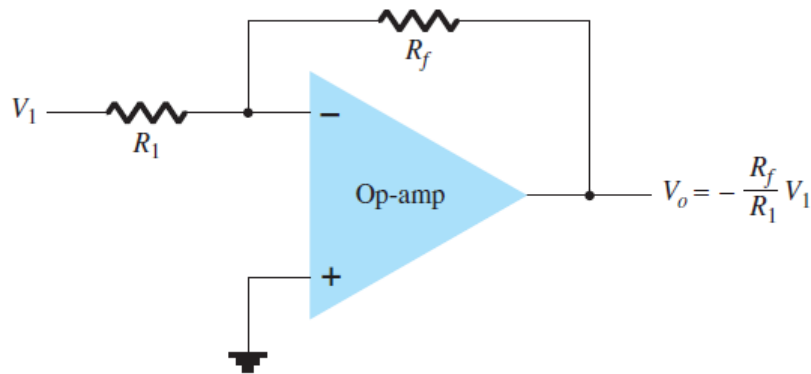


Figure 11.3.1: Inverting constant-gain multiplier.

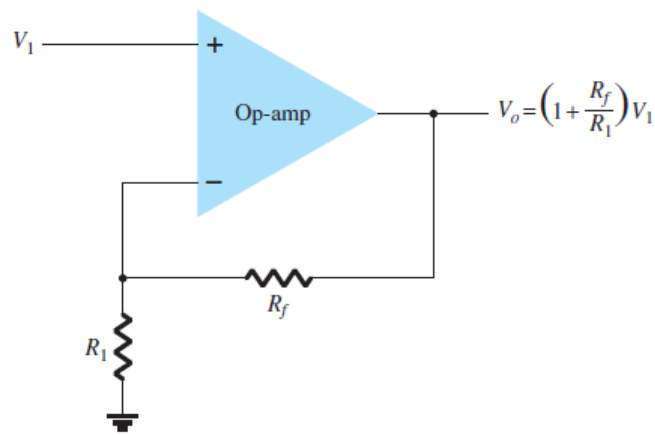


Figure 11.3.2: Non-Inverting constant-gain multiplier.

11.4 Activity 1 : To design an inverting amplifier with a gain of 200 with a sinusoidal input of 2.5 mV.

11.4.1 Procedure

1. Connect the circuit as shown in Fig. 11.3.1 with certain modifications.
2. Determine the values of resistor needed to get a gain of 100.
3. This circuit must be connected as per the pin configuration of the IC. Refer to Fig. 11.3.3.
4. Apply a frequency of 10 KHz with the given magnitude of 2.5 mV.
5. Observe both input and outputs on both the channels of DSO.
6. Verify whether the gain obtained experimentally is close to the theoretical value or not.

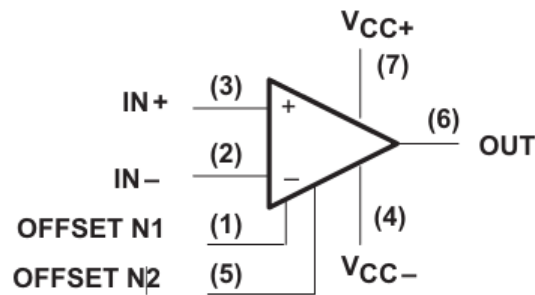


Figure 11.3.3: Pin Configuration of Op-741

11.5 Activity 2 : To design an non-inverting amplifier with a gain of close to 110 with a sinusoidal input of 2.5 mV.

11.5.1 Procedure

1. Connect the circuit as shown in Fig. 11.3.2 with certain modifications.
2. Determine the values of resistor needed to get a gain of 110.
3. This circuit must be connected as per the pin configuration of the IC. Refer to Fig. 11.3.3.
4. Apply a frequency of 10 KHz with the given magnitude of 2.5 mV.
5. Observe both input and outputs on both the channels of DSO.
6. Verify whether the gain obtained experimentally is close to the theoretical value or not.

11.5.2 Conclusion:

Experiment Number 12

Aim: To realize adder and subtractor using Op-Amp IC $\mu A741$.

12.1 Equipment/Components required

Dual power supply, Function generator, DSO, Bread board, IC -741, Resistors, Probes and connecting wires as required.

12.2 Theory

The Summing Amplifier, Fig. 12.2.1, is another type of operational amplifier circuit configuration that is used to combine the voltages present on two or more inputs into a single output voltage.

The differential amplifier, Fig. 12.2.2, amplifies the voltage difference present on its inverting and non-inverting inputs. The differential amplifier is a voltage subtractor circuit which produces an output voltage proportional to the voltage difference of two input signals applied to the inputs of the inverting and non-inverting terminals of an operational amplifier.

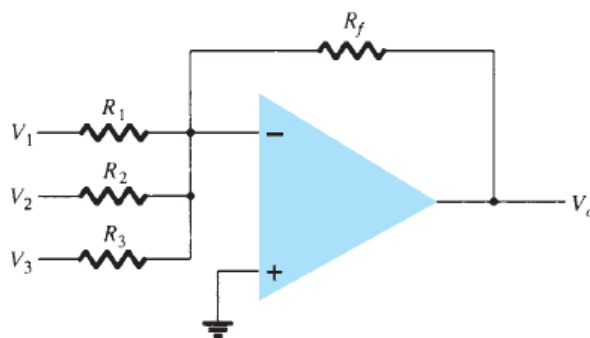


Figure 12.2.1: Summing amplifier

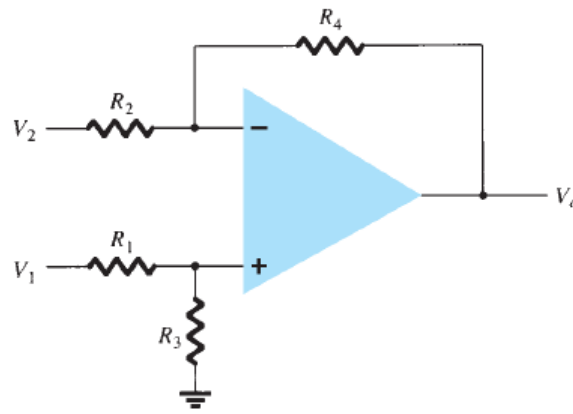


Figure 12.2.2: Difference amplifier

12.3 Activity 1 : To determine the output voltage for two input summing amplifier. The inputs are $V_1 = 50 \sin(1000t)$ mV and $V_2 = 10 \sin(3000t)$ mV.

12.3.1 Procedure

1. Connect the circuit as shown in Fig. 12.2.1 with certain modifications.
2. Choose the values of $R_f = 330k\Omega$, $R_1 = 33k\Omega$ and $R_2 = 10k\Omega$
3. This circuit must be connected as per the pin configuration of the IC.
4. Apply a two inputs as per requirement. Note the difference in frequencies.
5. Observe both input and outputs on both the channels of DSO.
6. Verify whether the overall gain obtained experimentally is close to the theoretical value or not.

12.4 Activity 2 : To determine the output voltage for the difference amplifier. The inputs are $V_1 = 50 \sin(1000t)$ mV and $V_2 = 10 \sin(3000t)$ mV.

12.4.1 Procedure

1. Connect the circuit as shown in Fig. 12.2.2 with certain modifications.
2. Choose the values of $R_4 = 100k\Omega$, $R_2 = 100k\Omega$, $R_1 = 20k\Omega$, and $R_3 = 20k\Omega$
3. This circuit must be connected as per the pin configuration of the IC.
4. Apply a two inputs as per requirement. Or choose as per your convenience.

5. Observe both input and outputs on both the channels of DSO.
6. Verify whether the overall gain obtained experimentally is close to the theoretical value or not.

12.4.2 Conclusion:

Experiment Number 13

BJT Common Base Characteristics

13.1 Aim: To study the BJT Common Base Characteristics

13.2 Link required

<https://be-iitkgp.vlabs.ac.in/exp/common-base-characteristics/>

13.3 Theory

In BJT common base (CB) configuration, the base terminal is common to both the input (emitter) and output (collector) circuits. The input signal is applied between the emitter and base, while the output is taken from the collector and base. This configuration is characterized by low input impedance and high output impedance, with a current gain less than unity.

13.4 Activity

1. GO to the link and perform the experiment as per the instructions.
2. Take printouts of the results and attach them in your file/record.

Experiment Number 14

Study of basic properties of operational amplifier

14.1 Aim: To study the basic properties of operational amplifier: inverting and non-inverting amplifiers

14.2 Link required

<https://be-iitkgp.vlabs.ac.in/exp/non-inverting-amplifiers/theory.html>

14.3 Theory

Operational Amplifier commonly known as Op-Amp, is a linear electronic device having three terminals, two high impedance input and one output terminal. Op-Amp can perform multiple function when attached to different feedback combinations like resistive, capacitive or both. Generally it is used as voltage amplifier and the output voltage of the Op-Amp is the difference between the voltages at its two input terminals.

14.4 Activity

1. GO to the link and perform the experiment as per the instructions.
2. Take printouts of the results and attach them in your file/record.